AI-Based News Summarizer Using Natural Language Processing

Submitted in partial fulfillment of the requirements of the
Mini-Project of **Natural Language Processing Lab** for Fourth Year of

Bachelors of Engineering
by

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CERTIFICATE

This is to certify that the mini-project entitled **"AI-Based News Summarizer Using Natural Language Processing"** is a bonafide work of **Khan Yahya Shahid Ali – 28** submitted to the University of Mumbai in partial fulfillment of the requirement for the Mini-Project of **Natural Language Processing Lab** for Fourth Year of the Bachelor of Engineering in **“Artificial Intelligence & Data Science”**.

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Declaration

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

NewsBrief AI is an interactive, web-based Natural Language Processing (NLP) application developed using Python and the Streamlit framework. The primary objective of this project is to automate the summarization of lengthy textual news articles, reports, and documentation, condensing extensive narratives into concise executive summaries and bullet points without losing critical informational context.

The system implements a Frequency-Weighted Extractive Summarization architecture. It segments input text into discrete sentence units, removes syntactic stop-words, and calculates normalized term-frequency weights for all vocabulary words. Each sentence is subsequently evaluated using a length-penalized salience scoring function: sentences possessing high cumulative keyword density are prioritized. The top-ranked sentences are chronologically re-ordered to ensure narrative coherence. Additionally, the system computes document statistics including original word count, summarized word count, compression reduction percentage (typically 55% to 70%), and estimated reading time saved.

NewsBrief AI operates 100% locally with zero external paid API dependencies or costly cloud inference costs. The application provides an intuitive web interface featuring dynamic compression sliders, 1-click preset articles across diverse domains (AI & Tech, Space, and Clean Energy), and dual display tabs for executive summaries and key bullet takeaways. Overall, the project demonstrates how classical NLP principles can be harnessed to build practical, real-world productivity tools.

INDEX

Sr. No.	Title	Page No.
1.	Aim	1
2.	Problem Statement	1
3.	Brief Theory	2
4.	Implementation Explanation	4
5.	Results	5
6.	Conclusion	6
7.	References	7

Chapter 1

Aim

The aim of the **AI-Based News Summarizer Using Natural Language Processing** project is to develop a robust, interactive web application using Python and the Streamlit framework that automatically condenses lengthy unstructured news articles into precise executive summaries and key bullet insights. The system aims to implement an extractive NLP pipeline that accurately calculates word salience weights, computes length-normalized sentence scores, extracts top key concepts, and presents comparative document statistics through an accessible, modern dashboard.

Chapter 2

Problem Statement

In modern digital media, the exponential growth of textual publications, editorial columns, and technical documentation creates severe information overload. Readers and knowledge workers spend hours parsing voluminous text to identify core factual takeaways. Manual reading and summarization are time-consuming, subjective, and difficult to scale across multi-page articles. Furthermore, many existing enterprise AI summarization services rely on expensive commercial APIs that require persistent internet connectivity and present potential data privacy concerns. To address this challenge, NewsBrief AI delivers a localized, rule-based extractive NLP summarizer that compresses news stories by 50% to 70% while guaranteeing that all extracted facts represent exact, unhallucinated source statements.

Chapter 3

Brief Theory

Text summarization in Natural Language Processing is broadly categorized into two fundamental paradigms: **Abstractive Summarization** and **Extractive Summarization**. Abstractive techniques utilize sequence-to-sequence neural networks to generate new paraphrased sentences; however, they require massive compute resources and carry substantial risks of hallucination. Conversely, Extractive Summarization identifies and extracts the most significant sentences directly from the original document, preserving factual fidelity.

The core mathematical formulation of NewsBrief AI relies on **Frequency-Weighted Salience Scoring**. The process begins with sentence boundary disambiguation using regular expressions that distinguish punctuation terminals from honorifics and decimals. Next, word tokens are filtered against a comprehensive stop-word corpus containing syntactic particles (e.g., *the, is, and, which*). For each remaining meaningful word token *w*, its normalized term weight is defined as:

$$\textbf{Weight}(w) = \textbf{Frequency}(w) / \textbf{MaxFrequency}$$

Where *MaxFrequency* represents the occurrence count of the most dominant keyword in the passage.

To score an individual candidate sentence *S*, the system computes the aggregate weight of its member vocabulary words. To prevent artificially inflated scores for disproportionately long sentences, a length normalization penalty is applied:

$$\textbf{Score}(S) = \sum [\textbf{Weight}(w)] / \sqrt{[\textbf{Length}(S)]}$$

Sentences exceeding the selected compression threshold are extracted and sorted back into their original narrative sequence. This ensures that the final executive output maintains natural chronological flow and contextual coherence.

Chapter 4

Implementation Explanation

The implementation of NewsBrief AI is structured modularly in Python, utilizing standard libraries (re, math, collections.Counter) alongside streamlit and pandas. The application logic is divided into three primary tiers: the Preprocessing Module, the Salience Scoring Engine, and the Streamlit GUI Dashboard.

1. Preprocessing & Tokenization: The split_into_sentences() function executes regex lookbehind splitting (?<=[.!?])\s+ to partition paragraphs without truncating abbreviations. Word tokenization extracts alpha-numeric character sequences converted to lowercase.

2. Weight Matrix & Sentence Scoring: In summarize_text(), words are cross-referenced with a 150-word stop-word set. The remaining tokens are tallied via Counter. Each sentence receives an objective salience score computed by summing normalized weights divided by the square root of word count.

3. Ranking & Assembly: The sentences are sorted by score, the top-*K* subset (governed by the user's ratio slider) is selected, and re-indexed to match document sequence.

4. Interactive Web Interface: Streamlit renders side-by-side split panels: the left panel hosts article text areas, dynamic compression sliders, and preset domain buttons; the right panel presents four metric cards (Original Words, Summary Words, Reduction %, Time Saved), executive summary boxes, key point bullet cards, and keyword frequency bar charts.

Chapter 5

Results

The AI-Based News Summarizer was tested extensively across news articles spanning multiple technical and societal domains, consistently achieving compression ratios between 55% and 65% without loss of key contextual concepts.



Figure 1: *User Interface and successful execution of NewsBrief AI Summarizer*

In the benchmark test on the *AI & Technology* article (Figure 1), the original text containing 109 words was successfully compressed to 40 words, achieving a **63% reduction** in reading overhead while capturing the central theme regarding autonomous agent workflows and engineering safety mandates. Keywords such as *autonomous*, *agents*, and *reasoning* were correctly identified as dominant entities.

Chapter 6

Conclusion

The **AI-Based News Summarizer Using Natural Language Processing** successfully achieves its core objective of automating document condensation through localized, frequency-weighted extractive NLP algorithms. The project eliminates the complexity and cost of third-party cloud APIs, delivering instantaneous executive summaries, structured bullet points, and quantitative reading metrics. While rule-based extractive algorithms do not generate novel paraphrased sentences, they offer complete protection against generative hallucinations, making them exceptionally reliable for factual news digestion. In future revisions, the platform can be augmented with lightweight transformer models (e.g., T5 or BART) for hybrid abstractive-extractive generation and multilingual Indian language translation.

Chapter 7

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